GPU AND COOLING SYSTEM

1.Open your own laptop or desktop (or use an online PC hardware simulator if you don't have access to a real system) and identify the GPU model installed. Write down the GPU's brand, model number, and whether it is integrated or dedicated.

=>Press **Windows + X**

Click **Device Manager**

Expand **Display adapters**

Note the GPU name shown there

2. Take a clear photo or screenshot of the cooling system (fan, heatsink, or liquid cooling setup) inside your PC or from a PC hardware simulator. Label the main cooling components and briefly describe their function.

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Component	Function
CPU Fan	Blows air over the heatsink to remove heat from the CPU.
Heatsink	Metal fins that absorb and dissipate heat from the CPU.
Case Fan	Moves hot air out of the case and brings cool air in.
Radiator (Liquid Cooling)	Transfers heat from the coolant to the air.
Pump (Liquid Cooling)	Circulates coolant through the cooling loop.

2. Create a comparison table listing at least 3 differences between air cooling and liquid cooling systems for GPUs, including factors like cost, efficiency, and maintenance.

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Factor	Air Cooling	Liquid Cooling
Cost	Less expensive	More expensive
Cooling	Good for normal gaming and	Better for high-performance gaming

Efficiency	everyday use	and overclocking
Maintenance	Requires little maintenance	Requires more maintenance and monitoring
Installation	Easy to install	More complex to install
Noise Level	Can be noisier due to fan speed	Usually quieter under heavy loads
Space Required	Takes up less space	Requires additional space for radiator and tubing

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Factor	Air Cooling	Liquid Cooling
Cost	Less expensive	More expensive
Cooling Efficiency	Good for normal gaming and everyday use	Better for high-performance gaming and overclocking
Maintenance	Requires little maintenance	Requires more maintenance and monitoring
Installation	Easy to install	More complex to install
Noise Level	Can be noisier due to fan speed	Usually quieter under heavy loads
Space Required	Takes up less space	Requires additional space for radiator and tubing

4. Imagine you are building a gaming PC for streaming IPL matches and playing graphics-heavy games. Choose between air cooling and liquid cooling for your GPU, and justify your choice in 3-4 sentences based on your research.

=>I would choose **liquid cooling** for my GPU when building a gaming PC for streaming IPL matches and playing graphics-heavy games. Liquid cooling is more efficient at removing heat, which helps keep the GPU temperature low during long gaming and streaming sessions. It also produces less noise compared to high-speed air-cooling fans. Although liquid cooling costs more and requires more maintenance, its superior cooling performance makes it the better option for a high-performance gaming PC.